Advanced SQL Queries Using Joins and Subqueries

Multi-table library reports with subqueries, grouping, and a CTE

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Project: Library Management System. **Output convention:** result tables below are reproducible expected outputs from the fixed rows in the shared SQL script. They are not screenshots of MySQL execution.

1. Current open-loan report using joins

The query joins loans to members, copies, and books so each circulation row shows a person, title, due date, and a status evaluated on the fixed date 2026-10-23.

```
SELECT l.loan_id, m.full_name, b.title, l.due_date,
       CASE WHEN l.due_date < DATE('2026-10-23')
            THEN 'OVERDUE' ELSE 'ON TIME' END AS loan_condition
FROM loans AS l
JOIN members AS m ON m.member_id = l.member_id
JOIN book_copies AS bc ON bc.copy_id = l.copy_id
JOIN books AS b ON b.book_id = bc.book_id
WHERE l.return_date IS NULL
ORDER BY l.due_date, l.loan_id;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

loan_id	full_name	title	due_date	loan_condition
1001	Asha Mehta	Relational Database Basics	2026-10-15	OVERDUE
1004	Asha Mehta	Python in Practice	2026-10-19	OVERDUE
1006	Sara Nair	Data Structures Made Clear	2026-10-22	OVERDUE
1002	Kabir Shah	Relational Database Basics	2026-10-24	ON TIME
1008	Sara Nair	Learning Web Design	2026-10-26	ON TIME

2. Books with no loan history using NOT EXISTS

```
SELECT b.book_id, b.title,
       SUM(CASE WHEN bc.copy_status = 'AVAILABLE' THEN 1 ELSE 0 END)
           AS available_copies
FROM books AS b
JOIN book_copies AS bc ON bc.book_id = b.book_id
WHERE NOT EXISTS (
    SELECT 1
    FROM book_copies AS bc2
    JOIN loans AS l ON l.copy_id = bc2.copy_id
    WHERE bc2.book_id = b.book_id
)
GROUP BY b.book_id, b.title
ORDER BY b.title;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

book_id	title	available_copies
8	Mystery at Lake Road	2

NOT EXISTS is a correlated subquery: a catalog book is returned only when there is no matching copy with a loan record.

3. Books loaned at least twice

```
SELECT b.title, COUNT(l.loan_id) AS loan_count
FROM books AS b
JOIN book_copies AS bc ON bc.book_id = b.book_id
JOIN loans AS l ON l.copy_id = bc.copy_id
GROUP BY b.book_id, b.title
HAVING COUNT(l.loan_id) >= 2
ORDER BY loan_count DESC, b.title;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

title	loan_count
Relational Database Basics	2

4. Members above the average loan count

```
WITH member_totals AS (
    SELECT m.member_id, m.full_name, COUNT(l.loan_id) AS loan_count
    FROM members AS m
    LEFT JOIN loans AS l ON l.member_id = m.member_id
    GROUP BY m.member_id, m.full_name
)
SELECT full_name, loan_count
FROM member_totals
WHERE loan_count > (SELECT AVG(loan_count) FROM member_totals)
ORDER BY loan_count DESC, full_name;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

full_name	loan_count
Sara Nair	3
Asha Mehta	2
Kabir Shah	2

The CTE first includes every member with a LEFT JOIN, including Isha with zero loans. The average is 1.6 across five members, so Sara, Asha, and Kabir are above it.

5. Pending reservation report

```
SELECT r.reservation_id, m.full_name, b.title,
       r.reservation_date, r.reservation_status
FROM reservations AS r
JOIN members AS m ON m.member_id = r.member_id
JOIN books AS b ON b.book_id = r.book_id
WHERE r.reservation_status = 'PENDING'
ORDER BY r.reservation_date, r.reservation_id;
```

Expected output from the deterministic sample rows in Library_Management_System_MySQL_8.sql:

reservation_id	full_name	title	reservation_date	reservation_status
2001	Isha Rao	Mystery at Lake Road	2026-10-19	PENDING
2004	Asha Mehta	Mystery at Lake Road	2026-10-20	PENDING

Screenshot steps

Run each report query separately after loading the shared seed data. Capture the query and its Result Grid in MySQL Workbench. The open-loan query deliberately uses a literal as-of date; if you change sample dates, update the expected result table as well.